Invited Article: Broadband highly efficient dielectric metadevices for polarization control

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Metadevices based on dielectric nanostructured surfaces with both electric and magnetic Mie-type resonances have resulted in the best efficiency to date for functional flat optics with only one disadvantage: a narrow operational bandwidth. Here we experimentally demonstrate *broadband* transparent all-dielectric metasurfaces for highly efficient polarization manipulation. We utilize the *generalized Huygens principle*, with a superposition of the scattering contributions from several electric and magnetic multipolar modes of the constituent meta-atoms, to achieve destructive interference in reflection over a large spectral bandwidth. By employing this novel concept, we demonstrate reflectionless (~90% transmission) half-wave plates, quarter-wave plates, and vector beam *q*-plates that can operate across multiple telecom bands with ~99% polarization conversion efficiency. © *2016 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (http://creativecommons.org/licenses/by/4.0/).*
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The concept of metamaterials—artificial electromagnetic structures with unconventional properties composed of subwavelength constituent elements—has opened many new opportunities for the efficient manipulation of light propagation and a design of novel types of metadevices with unusual properties.[1] In particular, *metasurfaces*, ultra-thin nanostructured layers, emerged recently as novel metadevices capable of reshaping transmitted and reflected light. Metasurfaces are composed of resonant subwavelength elements that are distributed spatially across a two-dimensional surface. Due to the resonant scattering of the incoming light, each element can alter the phase and amplitude of the light field. Metasurfaces thus enable a variety of different functionalities realized at the subwavelength scale,[2,3] including controlled beam deflection, flat-lens focusing, and holography. As such, metasurfaces can replace the conventional free-space bulky components for wavefront and polarization control, imaging, and even on-chip integration.

While many designs and functionalities of metasurfaces have been suggested based on plasmonic planar structures,[2,3] most of these metasurfaces demonstrate low efficiencies in transmission due to losses in their metallic components. Indeed, the maximum efficiency reported so far does not exceed 10%,[4] and usually much lower efficiencies are reported (see Refs. 5 and 6 to cite a few). In contrast, *all-dielectric resonant metasurfaces* completely avoid absorption losses, and can drastically enhance the overall efficiency,[7–12] especially in the transmission regime. However, there exists a *serious issue* that prevents the immediate use of all-dielectric metasurfaces in practical planar metadevices: the fixed wavelength[11,13] or narrow spectral range[9] of operation. Broadband and highly efficient functionality was demonstrated for metasurfaces operating in reflection,[14] but to date it has not been achieved for transmissive devices. Recently, transmissive metasurfaces operating at several

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different frequencies[15,16] were demonstrated. Multiple frequency operation resolves the issue of narrow-band response for many specific applications, such as multi-colour imaging. However, for a number of technological implementations of metasurfaces, e.g., for polarization control of light, a continuous spectral range of operation is required, accompanied by clear design principles that can enable such broad bandwidth.

In this paper, we suggest and demonstrate a novel approach for the design of broadband all-dielectric metasurfaces by overlapping the scattering contributions of several multipoles (both electric and magnetic) of the constituent nanoscale meta-atoms, thus producing high-transmission polarization and phase control over a wide spectral bandwidth. We employ this approach to demonstrate planar metadevices for efficient polarisation control, including half- or quarter-wave plates and vector beam q-plates. Our metadevices demonstrate near-100% energy efficiency in full-wave numerical simulations. 100% efficiency of a single-layer functional device is an advantage of employing resonant nanostructure as this allows for impedance-matching with air, which cannot be achieved with single-layer non-resonant functional structures. In experiment, the transmission efficiencies reach 90% and polarization conversion efficiency of ~99% across several telecommunication bands. While experimental energy efficiency is lower than theoretically predicted due to fabrication imperfections, it is among the highest demonstrated efficiencies of the metadevices. When compared to conventional birefringent media our devices demonstrate effective values of birefringence that are at least three times greater, and they allow for transverse patterning with pixel density at least three orders of magnitude better than any available liquid-crystal array.

To investigate the general operation of high-efficiency ultra-thin transmissive metadevices, first we review the basic requirements of operation for phase and polarization control.

1. The device should provide high transmission, as illustrated in Fig. 1(a): $T \rightarrow 100\%$.
2. The device should provide 2π phase control of transmission, relative to the incident wave, thus covering the entire phase space [Fig. 1(b)]: $\varphi \in [0, 2\pi]$.
3. The device should provide π control of the phase difference between the transmission of two orthogonal polarizations [Fig. 1(c)]: $\varphi_x - \varphi_y \in [0, \pi]$.

The former two requirements provide high-transmission phase control, while the latter requirement is further necessary to provide polarization control. In conventional optics, the first requirement of broadband anti-reflection device can be fulfilled by employing multi-wave interference techniques from multiple stacked layers. However, satisfying all three requirements in a single layer metasurface over a broad bandwidth appears as an impossible task.

To tackle this problem, we must revisit the scattering of the constituent elements of the metadevice from the perspective of *multipolar interference*. We notice that light scattered from any arbitrary object can be decomposed into multipoles that have even or odd parity under spatial inversion

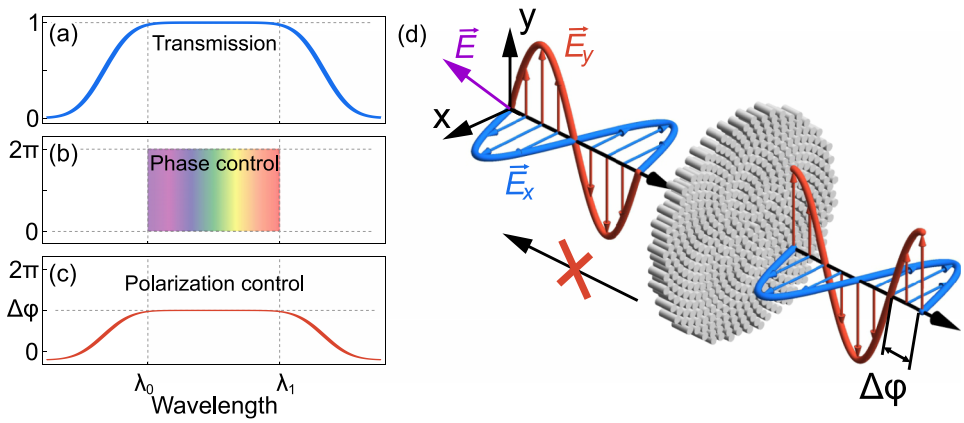


FIG. 1. Concept of a broadband highly efficient metadevice. (a) Near-unity energy efficiency over a broad spectral region. (b) Phase coverage in the range [0-2π]. (c) Phase difference for two orthogonal polarizations in the range [0-π]. (d) Schematic of a metadevice for spatially variant polarization control operating in a broad spectral range.

(see supplementary material[17] for the details of the multipolar decomposition), thereby allowing us to express the scattered field $\mathbf{E}$ as a sum of an even component $\mathbf{E}_{even}$ and an odd component $\mathbf{E}_{odd}$. As such, the transmission T and reflection R of a normally incident, co-polarized plane wave propagating in the $\hat{\mathbf{z}}$ direction, can be written as follows:

$$T = |1 + E_{even}(\hat{\mathbf{z}}) + E_{odd}(\hat{\mathbf{z}})|^2 , \tag{1}$$

$$R = |E_{even}(-\hat{\mathbf{z}}) + E_{odd}(-\hat{\mathbf{z}})|^2 = |E_{even}(\hat{\mathbf{z}}) - E_{odd}(\hat{\mathbf{z}})|^2 , \tag{2}$$

where we have defined the magnitude and phase of the incident field such that $E_{\mathrm{PW}}(\hat{\mathbf{z}}) = 1$, and have assumed negligible absorption. However, to further control the phase ϕ of reflectionless transmission, we need to satisfy

$$[1 + E_{even}(\hat{\mathbf{z}}) + E_{odd}(\hat{\mathbf{z}})] = e^{i\phi}. \tag{3}$$

Yet, because the magnitude is positive definite in Eq. (2), reflection can vanish only when $E_{even}(\hat{\mathbf{z}})$ and $E_{odd}(\hat{\mathbf{z}})$ are equal to each other in both magnitude and phase. As such, at any phase ϕ, the full transmission is achieved when the individual nanostructures of a metadevice radiate as

$$E_{even}(\hat{\mathbf{z}}) = E_{odd}(\hat{\mathbf{z}}) = \frac{1}{2}\left(e^{i\phi} - 1\right) . \tag{4}$$

Notably, this equation can be satisfied for a single wavelength with a minimum of two multipoles, each having opposite parity; such as an electric and a magnetic dipole resonance. This single-wavelength functionality has previously been demonstrated for Huygens' metasurfaces.[9,18,19] In particular, optical all-dielectric Huygens' metasurfaces rely on an overlap of the electric and magnetic resonances in high-index dielectric nanoparticles, resulting in zero backward scattering.[20,21] The idea of Huygens metasurface has proven successful in designing different functional metadevices, including vortex beam generators,[12,22] beam deflectors,[12,23] and holograms.[11,24] However, since the overlap of the electric and magnetic dipole resonances (the so-called Huygens' condition) occurs only in a narrow spectral range, it is believed that no broadband devices can be achieved in this way.

To satisfy Eq. (4) over a broad spectral range we instead employ a multipolar nanostructure, which makes E_{even} and E_{odd} each a linear combination of many multipoles. Importantly, Eq. (4) then becomes a balance between multipoles, and does not require resonances of equal amplitude and spectral width. In other words, *multipolar structures* are inherently suited for realizing broadband phase control. Additional discussion regarding this conclusion and its derivation is provided in the supplementary material.[17]

We can thereby achieve a broadband response as a result of multi-wave interference between the scattering waves produced by several pairs of multipoles of opposite parity. We refer to such multipolar forward scattering as the *generalized Huygens condition*, which resembles to an extent the forward scattering from two higher order multipoles called generalized Kerker condition.[25,26] Such multipolar scattering also resembles some principles of operation from multilayer dielectric coatings, where broadband functionality is a result of interference of multiple reflected waves.

We now focus on demonstrating the capacity of multipolar dielectric nanostructures for constructing broadband metasurfaces for highly efficient polarization control. Our metasurface consists of closely spaced silicon nanopillars, as shown in Fig. 2(a) to allow more multipolar modes in comparison with nanodisks. Here we assume negligible optical losses and focus on minimizing reflection. First, we relate the metasurface reflection to the magnitude of backward scattering from a single nanopillar with a superposition of individual contributions of higher-order multipoles. As seen in Fig. 2(b), the silicon nanopillar array approaches broadband near-unity transmission as we account for higher-order multipoles. In Fig. 2(c), we show that the origin of this broadband transmission is attributed to interference in the reflection between dipole- and higher-order multipoles (HOM). Both dipole-order multipoles and the higher-order multipoles can be seen to provide a significant contribution to reflection in isolation, while their sum suppresses reflection. Because of the negligible material losses of silicon, this broad suppression of reflection is sufficient to provide near-unity and broadband transmission through the metasurface.

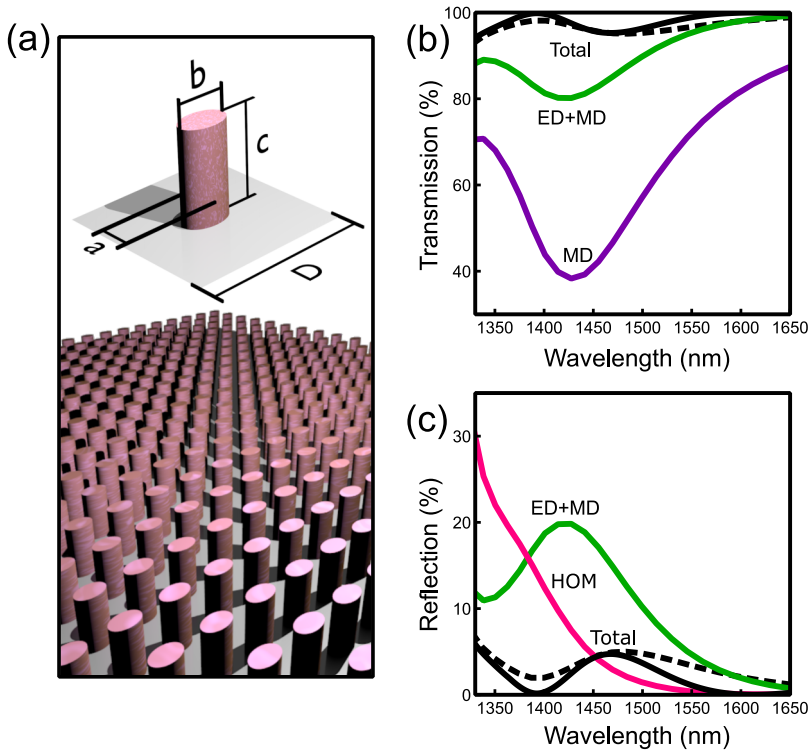


FIG. 2. (a) Schematic of our broadband metadevice. The spectra in (b) and (c) show the transmission and reflection of the metadevice and their dependence on the multipoles (ED = electric dipole, MD = magnetic dipole, and HOM = sum of higher-order multipoles). The black dashed curves in (b) and (c) show transmission and reflection from a complete set of multipoles, which neglects reflection off the glass substrate, and the black solid curves show the direct numerical transmission and reflection. Dimensions are $a = 285$ nm, $b = 450$ nm, $c = 860$ nm, and $D = 750$ nm, and the incident polarization is parallel to the short axis of the nanopillars.

As the next step, we need to achieve the control over the phase of the transmitted wave. In this regard, it is important to recognize that the forward scattering components from each active multipole must collectively suppress the incident field and construct a new wavefront that can have a different phase to that of the background field. Therefore, the scattering contributions from the various multipoles have to be non-negligible relative to the incident field. This is indeed the situation in Figs. 2(b) and 2(c), where both the dipole- and the higher-order multipoles are shown to provide a significant contribution to transmission and reflection in isolation.

To finally demonstrate the combined phase and polarization control of the transmitted wavefront, we can maintain a near constant relative phase difference between the transmitted wave polarized along each axis of the silicon nanopillars of Fig. 2(a). In particular, we consider the case of a half-wave plate in Fig. 3, and maintain a π phase difference between the transmission polarized along each axis. Importantly, our results show that the π phase difference is maintained together with near unity transmission over both C and L telecommunications bands.

High operation bandwidth is a result of multipolar interference that can be achieved in various systems not limited to dielectric nanopillars. However, similar nanopillar geometries do not necessarily meet the conditions for the multipolar interference, and thus may lead to essentially single-wavelength operation. In the supplementary material[17] we give an example of two metadevices with similar geometries of nanopillars, but featuring 8 times difference in the spectral bandwidth.

We then proceed to experimental realization and optical testing of the proposed broadband functionality. We fabricate our samples from polycrystalline silicon on quartz substrate using electron-beam lithography; see the supplementary material[17] for the fabrication details. Overall, we demonstrate *three types of optical metadevices*—a half-wave plate, a quarter-wave plate, and

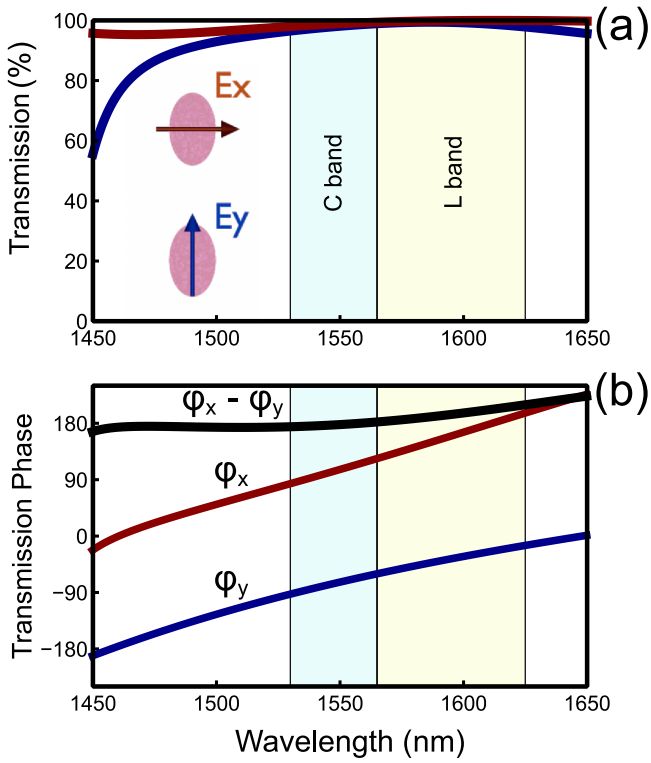


FIG. 3. (a) Transmission spectrum of the metadevice shown in Fig. 2, for polarizations parallel to the two axes of the nanopillar. (b) Phase of the transmitted waves with respect to the incident field for both orthogonal polarizations. The phase difference $\varphi_x - \varphi_y$ shows a constant value of π over a broad spectral range, as required for a broadband half-wave plate.

a q-plate all working in the telecom spectral region. The half-wave plate is an element that adds π phase difference between two principal linear polarizations of light. It is used for polarization rotation by upto 90°. Our half-wave plate consists of nanoparticles with sizes $a = 280$ nm, $b = 430$ nm, $c = 850$ nm, arranged into a square lattice with period $D = 750$ nm. The q-plates are used for conversion of linearly polarized beam into radially, or azimuthally polarized. Our q-plate consists of the same nanoparticles arranged into hexagonal lattice with the same density of particles per unit area. The spatial orientation of the nanoparticles, however varies spatially across the q-plate, as shown in Fig. 5. The quarter-wave plate is an element that adds $\pi/2$ phase difference between two principal linear polarizations of light. It is used for polarization conversions between linear, circular, and intermediate elliptical polarizations. Our quarter-wave plate consists of nanoparticles with sizes $a = 310$ nm, $b = 370$ nm, same height $c = 850$ nm and arranged into the same square lattice ($D = 750$ nm).

We then study experimentally the optical properties of the fabricated wave-plates. First, we measure their transmission spectra. For this, we employ a home-built white-light spectrometer that uses a tungsten light bulb as a broadband optical source and an optical spectrum analyzer detection. The transmission spectra of the samples are shown in Figs. 4(a) and 4(c)–black lines.

To check the basic functionalities of the metadevices, we measure how well the half-wave plate can rotate linear polarization by 90° and how well the quarter-wave plate can convert linear polarization into a circular one. We start with the fabricated half-wave plate and place it between a linear polarizer and analyzer with the wave plate axis being at 45° to the axis of the polarizer (the wave plate axis is defined here as the long axis of constituent elliptical nanoparticles). We observe that, in this case, the half-wave plate rotates the linear polarization of virtually 100% of transmitted light by 90° over a broad spectral region (shown as shaded region in the plots). In particular, Fig. 4(a) shows that transmission through a wave plate positioned between two crossed polarizers is reaching its maximum value, and transmission through a wave plate positioned between two parallel polarizers almost vanishes.

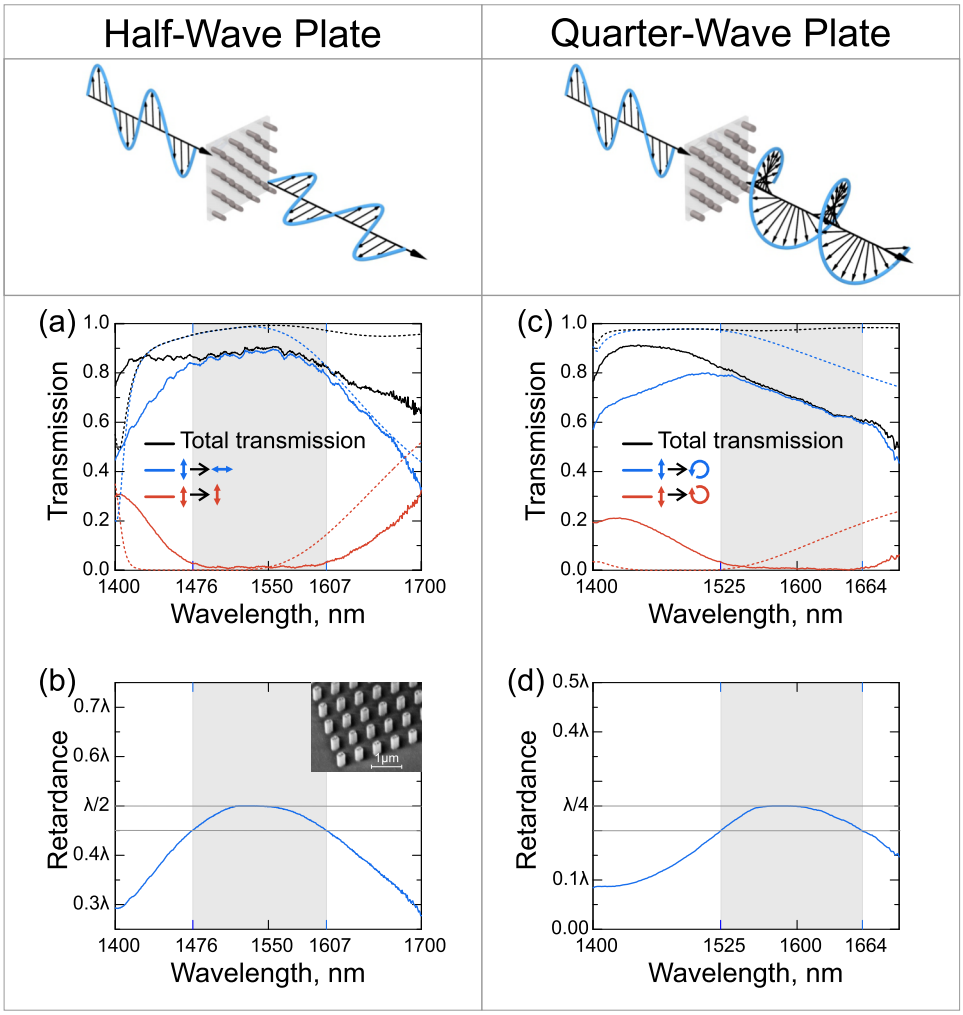


FIG. 4. Experimental demonstration of half- and quarter-wave plates. (a) Experimental and theoretical transmission spectra of the half-wave plate. Black line—*total transmission*. Red line—transmission through a metasurface positioned between a pair of linear *parallel polarizer and analyzer* with polarization axis oriented at 45° to metasurface's anisotropy axis. Blue line—transmission through a metasurface positioned between a pair of linear *crossed polarizer and analyzer* with polarization axis oriented at ±45° to metasurface's anisotropy axis. Solid lines—experimental data, dashed lines—theoretical results. (b) Experimentally retrieved retardance of the half-wave plate. Highlighted region shows the spectral range, with a deviation from $\lambda/2$ by no more that 10%, or $\lambda/20$. Inset shows a scanning electron micrograph of the fabricated half-wave plate. (c) Experimental and theoretical transmission spectra of the quarter-wave plate. Black line—*total transmission*. Red line—transmission through a metasurface positioned between a pair of linear polarizer with polarization axis oriented at 45° to metasurface's anisotropy axis and a right-circular analyzer. Blue line—transmission through a pair of linear polarizer with polarization axis oriented at 45° to metasurface's anisotropy axis and a left-circular analyzer. Solid lines—experimental data, dashed lines—theoretical results. (d) Experimentally retrieved retardance of the quarter-wave plate. Highlighted region show the spectral range, with retardance deviation from $\lambda/4$ by no more than $\lambda/20$.

Similarly, we test the performance of the quarter-wave plate. We place a linear polarizer in front of it and orient the wave plate axis at 45° with respect to the polarizer's axis. After the wave plate we place a circular polarizer (transmitting either left- or right-circular polarization) made of a superachromatic quarter-wave plate (from B. Halle) and a linear analyzer. We demonstrate virtually 100% conversion of linear polarization into circular over a broad spectral region, as shown in Fig. 4(c).

We further perform the full polarimetry measurements of the transmission spectra of the metasurfaces. For this we keep a vertically aligned linear polarizer and orient the samples anisotropy axis at 45°. We use a set of six different analyzers (six different orientations of the super-achromatic

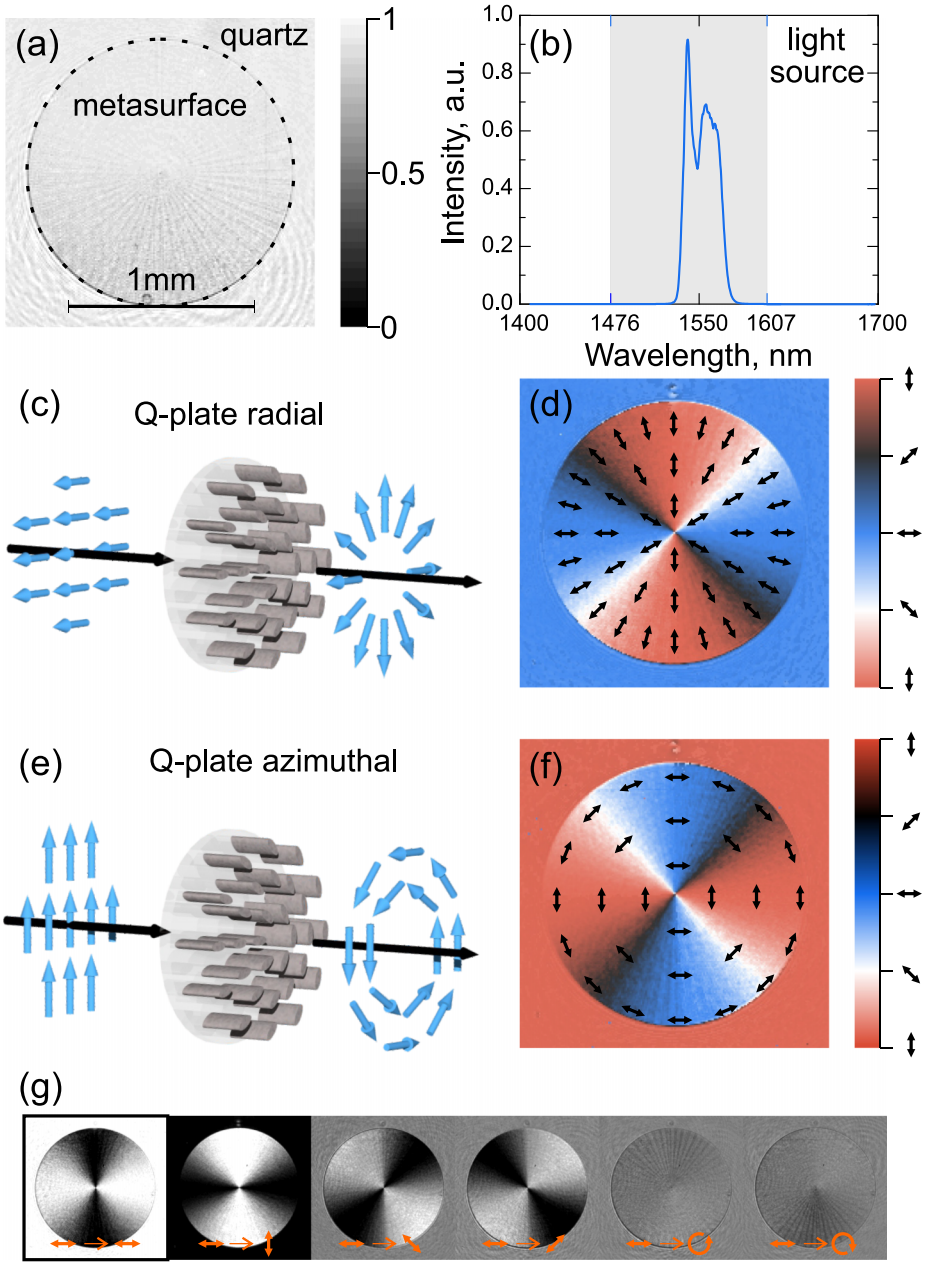


FIG. 5. Experimental demonstration of highly efficient q-plates. (a) Transmission image of the q-plate 1.5 mm in diameter placed on top of a quartz wafer, taken with an amplified spontaneous emission (ASE) broadband light source at around 1550 nm wavelengths. Note that the device appears to be nearly transparent. (b) Spectrum of the ASE light source used for the experiment. (c) q-plate functionality concept: conversion of vertical into azimuthal polarization. (d) Experimentally measured spatially resolved polarization of the azimuthally polarized output beam. (e) The q-plate functionality concept: conversion of horizontal into radial polarization. (f) Experimentally measured spatially resolved polarization of the radially polarized output beam. (g) Examples of light transmission through the q-plate positioned between different polarizers and analyzers (polarizer axes orientations marked in insets).

quarter-wave plate and a linear analyzer): linear horizontal and vertical, two linear diagonal, left-circular and right-circular analyzers; and correspondingly record six spectra. We then use the Stokes vector formalism to retrieve the full information about the polarization states of the transmission spectra in terms of their polarization degree, ellipticity and polarization rotation angle; see supplementary material[17] for more details on Stokes formalism. The extracted retardance of our

wave plates is shown in Figs. 4(b) and 4(d). Clearly, the measured retardance is close to 0.5 for the case of the half-wave plate and 0.25 for the case of the quarter-wave plate. Importantly, these values are constant (within the practically accepted tolerances) over a broad spectral range.

The biggest significance of the metasurface approach however comes from the ability to spatially vary the degree of polarization rotation. We therefore demonstrate experimentally the broadband operation of a q-plate, which is a device that converts beam with linear vertical polarization into a beam with azimuthal polarization and a beam with horizontal polarization into a beam with radial polarization [see Fig. 5]. We build an image of the q-plate on an infrared Xenics camera and use an amplified spontaneous emission broadband light source such that on the camera we receive an image [see Fig. 5(a)] *integrated* over the entire spectral range of our source, shown in Fig. 5(b). The q-plate shows ~90% overall transmission and it looks nearly transparent on the camera. Similarly to the studies of the wave plates, we perform the full polarimetry of the optical images (see more details in supplementary material[17]), i.e., we capture six images of the q-plate when the light passes through six different analyzers [see Fig. 5(g)]. We then employ the Stokes vector formalism and, in particular, retrieve polarization angles of light that passes through different spatial points of the q-plate [see Figs. 5(d) and 5(f), different colors correspond to different directions of linear polarization]. Figures 5(d) and 5(f) show that the metadevice allows virtually perfect conversion of vertical polarization into radial polarization and horizontal polarization into azimuthal polarization.

In conclusion, we have designed and demonstrated transparent all-dielectric metadevices for highly efficient polarization manipulation based on the generalized Huygens' principle, including half-wave plates, quarter-wave plates, and vector beam q-plates that can operate across several telecom bands. While the demonstrated metadevices allow to change the polarization of light in a manner similar to conventional birefringent crystals and liquid crystals, the effective value of birefringence is much higher than that in conventional materials. Indeed, for both half-wave plates and q-plates, the π phase difference is accumulated over 850 nm distance at 1550 nm wavelength, being equivalent to $\Delta n = (n_o - n_e) \approx 0.9$. The highest values of Δn for conventional materials are of the order of 0.3.[27,28] The metadevices feature subdiffractional spatial resolution of parameters variation, which is three orders of magnitude higher than that in the state-of-the-art liquid crystal arrays. The spectral bandwidth of the demonstrated metadevices is comparable or larger than the bandwidth of conventional zero-order wave retarders and diffractive optical elements.

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Supplemental Material: Broadband highly-efficient dielectric metasurfaces for polarization control

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A. BROADBAND HUYGEN'S METASURFACES

We begin by acknowledging that it is always possible to express the radiation from an arbitrary compact source as the sum of two components: one that is even (unchanged) under spatial inversion, and one that is odd (flips sign). This can be shown by considering the inversion behavior of spherical harmonics Y_{lm}:

$$Y_{lm}(\pi-\theta,\phi+\pi) = (-1)^l\, Y_{lm}(\theta,\phi)\ , \tag{1}$$

where θ is the polar angle of a spherical coordinate system and ϕ is the azimuthal angle. Any scattered electric field $\mathbf{E}$ can be written as a spherical multipole expansion[1,2] using vector spherical harmonics $\mathbf{X}_{lm}$, which are proportional to $(\mathbf{r}\times\nabla)\,Y_{lm}$, making each $\mathbf{X}_{lm}$ have the same parity as its associated Y_{lm}.

$$\mathbf{E}(\mathbf{r}) = \sum_{l=1}^{\infty}\sum_{m=-l}^{l} i^l\sqrt{\pi(2l+1)}\Big[\frac{a_{lm}}{k}\nabla\times(h_l^{(1)}(kr)\,\mathbf{X}_{lm}) + b_{lm}h_l^{(1)}(kr)\,\mathbf{X}_{lm}\Big]\ . \tag{2}$$

Here, $\mathbf{r}$ is the observation position vector, k is the wavenumber and $h_l^{(1)}$ is a spherical Hankel function of the first kind. As seen through this multipole expansion, the scattered field $\mathbf{E}$ can be written as the sum of two groups of multipoles that respectively have even and odd parity under spatial inversion. We will refer to these even and odd components of $\mathbf{E}$ as $\mathbf{E}_e$ and $\mathbf{E}_o$. More explicitly, using Equations (1) and (2), $\mathbf{E}_e$ will contain a_{lm} multipoles with odd values of l, and b_{lm} multipoles with even values of l; while $\mathbf{E}_o$ will contain the remaining a_{lm} and b_{lm} multipoles with even and odd values of l, respectively. If we now consider a sufficiently dense metasurface array, we can calculate the fields scattered by an individual structure in the forward ($\hat{\mathbf{z}}$) and backward ($-\hat{\mathbf{z}}$) directions and these values determine transmission and reflection of the array. Such forward and backward scattered fields can, however, be expressed in terms of their even and odd components:

$$\mathbf{E}(\hat{\mathbf{z}}) = \mathbf{E}_e(\hat{\mathbf{z}}) + \mathbf{E}_o(\hat{\mathbf{z}}) \tag{3}$$

$$\mathbf{E}(-\hat{\mathbf{z}}) = \mathbf{E}_e(\hat{\mathbf{z}}) - \mathbf{E}_o(\hat{\mathbf{z}}) \tag{4}$$

For simpler notation, we can consider the case for only one polarization in forward and backward directions, which allows us to write the following expression for the transmission T and reflection R of a normally-incident, co-polarized plane wave propagating in the $\hat{\mathbf{z}}$ direction.

$$T = |1 + E_e(\hat{\mathbf{z}}) + E_o(\hat{\mathbf{z}})|^2\ , \tag{5}$$

$$R = |E_e(-\hat{\mathbf{z}}) + E_o(-\hat{\mathbf{z}})|^2 = |E_e(\hat{\mathbf{z}}) - E_o(\hat{\mathbf{z}})|^2\ . \tag{6}$$

where we have defined the incident field such that $E_i(\hat{\mathbf{z}}) = 1$. To then control the phase ϕ of transmission relative to the plane wave, and with no loss, we also need to satisfy:

$$1 + E_e(\hat{\mathbf{z}}) + E_o(\hat{\mathbf{z}}) = e^{i\phi}\ . \tag{7}$$

However, refering to Equation (6), magnitude is positive definite and, therefore, reflection can only go to zero when $E_e(\hat{\mathbf{z}})$ and $E_o(\hat{\mathbf{z}})$ are equal to each other in both magnitude and phase. As such, full transmission at any phase ϕ is achieved when the indivdual nanostructures of a metasurface radiate as:

$$E_e(\hat{\mathbf{z}}) = E_o(\hat{\mathbf{z}}) = \frac{e^{i\phi}-1}{2}\ . \tag{8}$$

Existing realizations of Huygen's metasurfaces utilize both electric and magnetic dipole moments of individual nanostructures to play the role of the even and odd scattering components, respectively. These metasurfaces offer a conceptually simple system, but they have limited capacity for broadband responses because the only condition by which an electric dipole resonance and a magnetic dipole resonance can satisfy Equation (8) is if these two resonances cross[3,4]. Such crossings inherently occur at single wavelengths and make the approach relatively impractical for producing broadband responses. If we instead utilize a generalized Huygen's geometry that exhibits a large number of multipoles, $\mathbf{E}_e$ and $\mathbf{E}_o$ will each be described by a linear combination of several multipoles, each of which are then controlled independently by a combination of design parameters, at least in the ideal case. The key idea here is that Equation (8) can be realized without a crossing between two resonances, because it can instead be realized as a balance between many multipoles; an approach which is more amenable for realizations of over broad bandwidths.

We can now finish by briefly examining the distinction between dipolar and multipolar Huygen's metadevices at a more fundamental level. In particular, we can consider the underlying eigenmodes and eigenvalues associated with each individual resonance[5,6]. The radiation pattern of any given electric or magnetic dipole is static, hence resonances associated with solely an electric or magnetic dipole will always radiate in the same way, irrespective to the number of design parameters we change. The effect of tailoring parameters in a dipolar Huygen's metadevice will therefore only change each resonance's magnitude and phase, that is: its *eigenvalue*. On the other hand, if we have many multipoles present in the individual unit structures, a resonance will generally be associated with some linear combination of many multipoles, where this linear combination will be dependent on design parameters. Therefore, design parameters will also be able to change a resonance's radiation pattern, its *eigenmode*, in addition to the eigenvalue. To relate this new dimension of design control back to our previous discussion on scattering in the forward and backward direction, we can simply acknowledge that the radiative efficiency in the $\hat{\mathbf{z}}$ or $-\hat{\mathbf{z}}$ direction is different for each multipole [refer to Equation (2)]. As such, a difference in radiative efficiency of two resonances allows Equation (8) to be satisified even when there is no crossing of resonances. We have thereby reached a conclusion equivalent to the one made earlier, but have been able to get some insight with respect to resonances themselves.

B. SPHERICAL MULTIPOLE ANALYSIS FOR ARRAYS

Here we provide a derivation of the relationship between the forward and backward scattered fields from an array to that coming from a single unit cell in the array. Performing then a spherical multipole decomposition of the scattered field from a single element in the array, this relationship will allow us to obtain the corresponding multipole decomposition of forward and backward scattering from the array, and subsequently that of the transmission and reflection. The approach presented here is used to obtain the multipole decomposition transmission and reflection from full wave numerics, which is shown in Figure 2 of the main text. This section is based on the approach and derivations of Evlyukhin *et al.*[7], which we generalize to be in terms of induced currents rather than dipole moments.

We begin by considering the current distribution $\mathbf{J}$, which consists of both free and polarization currents, induced in a single unit cell of an infinite array. The electric field $\mathbf{E}$ radiated by this current distribution, external to the scattering object[8], can then be written in terms of Green's functions $\bar{G}_0$:

$$\mathbf{E}(\mathbf{r}) = i\omega\mu_0 \int_V \bar{G}_0(\mathbf{r},\mathbf{r}')\mathbf{J}(\mathbf{r}')\ \mathrm{d}r'^3 \tag{9}$$

where k is the wavenumber, ω is the angular frequency, V is the volume of the scattering object, which we assume here to be finite. The remaining part of the expression is the free space dyadic Green's function $\bar{G}_0$:

$$\bar{G}_0(\mathbf{r},\mathbf{r}') = \left[\bar{\mathrm{I}} + \frac{1}{k^2}\boldsymbol{\nabla}\boldsymbol{\nabla}\right]\frac{e^{ikR}}{4\pi R} \tag{10}$$

$$= \left[\Big(1 + \frac{i}{kR} - \frac{1}{k^2R^2}\Big)\bar{I} - \Big(1 + \frac{3i}{kR} - \frac{3}{k^2R^2}\Big)\hat{\mathbf{R}}\hat{\mathbf{R}}\right]\frac{e^{ikR}}{4\pi R}\,, \tag{11}$$

where $R = |\mathbf{r} - \mathbf{r}'|$, $\hat{\mathbf{R}}$ is the unit vector pointing from $\mathbf{r}'$ to $\mathbf{r}$ and $\bar{I}$ is the identity matrix. If we then consider the scattered eletric field in the far field ($R \to \infty$), the expression simplifies to:

$$\mathbf{E}(\mathbf{r}) \approx i\omega\mu_0 \int_V \frac{e^{ikR}}{4\pi R}\mathbf{J}_\perp(\mathbf{r}')\ \mathrm{d}r'^3 \tag{12}$$

where $\mathbf{J}_\perp$ contains only the components of $\mathbf{J}$ perpendicular to $\hat{\mathbf{R}}$. For the electric far field in specifically the forward ($\hat{\mathbf{z}}$) or backward ($-\hat{\mathbf{z}}$) scattering direction, we can make the approximation that $R \approx |\mathbf{z}|$, and keep only the leading order of $|\mathbf{z}|$ to get the following expression:

$$\mathbf{E}(\pm\mathbf{z}) \approx i\omega\mu_0 \frac{e^{ik|\mathbf{z}|}}{4\pi|\mathbf{z}|} \int_V \mathbf{J_{xy}}(\mathbf{r}') e^{ik(|\mathbf{z}-\mathbf{r}'|-|\mathbf{z}|)}\, \mathrm{d}r'^3 \,, \tag{13}$$

where $\mathbf{J_{xy}}$ contains only the x and y components of $\mathbf{J}$. To now extend this approach to account for a square array with unit cell period D, we can acknowledge that, for a normally-incident plane wave, the currents induced in each unit cell are related:

$$\mathbf{J}(\mathbf{r} + (p\hat{\mathbf{x}} + q\hat{\mathbf{y}})D) = \mathbf{J}(\mathbf{r}) \,, \tag{14}$$

where p and q are integers, and $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ denote unit vectors parallel to the periodic axes of the surface. Using this relation, we can now produce an expression for the electric field $\mathbf{E}_A$ radiated by the infinite array:

$$\mathbf{E}_A(\mathbf{r}) = \sum_{p,q=-\infty}^{\infty} i\omega\mu_0 \int_V \bar{G}_0\big(\mathbf{r}, \mathbf{r}' + (p\hat{\mathbf{x}} + q\hat{\mathbf{y}})D\big)\mathbf{J}\big(\mathbf{r}' + (p\hat{\mathbf{x}} + q\hat{\mathbf{y}})D\big)\, \mathrm{d}r'^3 \tag{15}$$

$$= i\omega\mu_0 \int_V \underbrace{\Big[\sum_{p,q=-\infty}^{\infty} \bar{G}_0\big(\mathbf{r}, \mathbf{r}' + (p\hat{\mathbf{x}} + q\hat{\mathbf{y}})D\big)\Big]}_{\equiv\, \bar{\mathcal{G}}(\mathbf{r}, \mathbf{r}')} \mathbf{J}(\mathbf{r}')\, \mathrm{d}r'^3 \,. \tag{16}$$

Evlyukhin *et al.*[7] showed that a square lattice makes the xy and yx off-diagonal elements of $\bar{\mathcal{G}}$ zero, and then went further to derive an expression for the diagonal terms:

$$\mathcal{G}_{xx}(\mathbf{r}, \mathbf{r}') = \mathcal{G}_{yy}(\mathbf{r}, \mathbf{r}') = \frac{i}{2k^2D^2} \sum_{\mathbf{L}}^{\infty} \frac{k^2 - L_x^2}{\sqrt{k^2 - \mathbf{L}^2}}\, e^{-i\mathbf{L}\mathbf{r}_\parallel + i|\mathbf{z}|\sqrt{k^2 - L^2}} \,, \tag{17}$$

where $\mathbf{r}_\parallel$ is the component of $\mathbf{r}$ parallel to the surface, and $\mathbf{L}$ are the reciprocal-lattice vectors of the array. For our purposes, we can use the simplified version of this expression for subwavelength lattice spacing and far field scattering. Moreover, because all terms other than $\mathbf{L} = \mathbf{0}$ are evanescent when we have a subwavelength lattice period D, the expression for the far field becomes much simpler:

$$\mathcal{G}_{xx}(\mathbf{r}, \mathbf{0}) \approx \frac{ie^{ikR}}{2kD^2} \,. \tag{18}$$

Therefore, by combining Equations (16) and (18), the farfield scattering in the forward and backward directions from the array is given by:

$$\mathbf{E}_A(\pm\mathbf{z}) = i\omega\mu_0 \frac{ike^{ik|\mathbf{z}|}}{2D^2} \int_V \mathbf{J_{xy}}(\mathbf{r}') e^{ik(|\mathbf{z}-\mathbf{r}'|-|\mathbf{z}|)}\, \mathrm{d}r'^3 \,. \tag{19}$$

If we then substitute Equation (13) into Equation (19), we are able to express the scattered field from an array in terms of the scattered field from the individual unit cell.

$$\mathbf{E}_A(\pm\mathbf{z}) = \frac{4\pi ik|\hat{\mathbf{z}}|}{2D^2} e^{ik|\mathbf{z}|} e^{-ik|\hat{\mathbf{z}}|} \mathbf{E}(\pm\hat{\mathbf{z}}) \tag{20}$$

$$= \left(\frac{\lambda}{D^2}\right) ie^{ik(|\mathbf{z}|-1)} \mathbf{E}(\pm\hat{\mathbf{z}}) \,. \tag{21}$$

Note that we have used $|\hat{\mathbf{z}}| = 1$ m in this expression, and this balances the units in the final expression. From this, we are able to calculate the reflection of a unit amplitude plane wave as:

$$R = \left(\frac{\lambda}{D^2}\right)^2 |\mathbf{E}(-\hat{\mathbf{z}})|^2 \tag{22}$$

If we then have negligble absorption in the unit cell, we can write the transmission as:

$$T = 1 - R \tag{23}$$

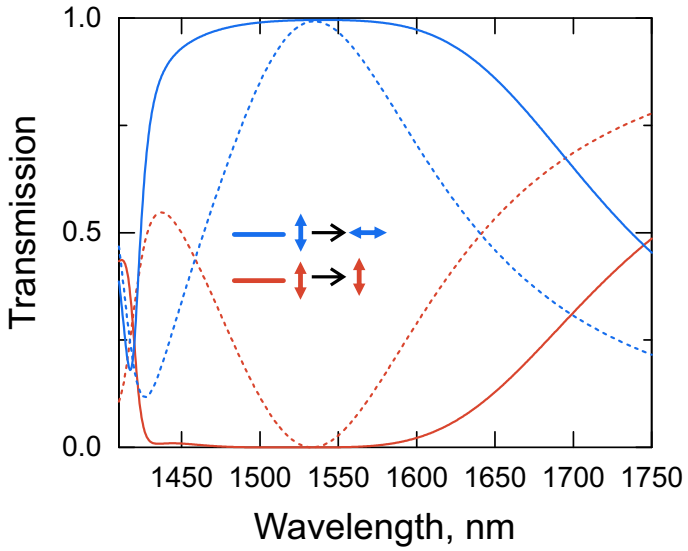


FIG. S1. **Broadband and narrowband waveplate metadevices based on Si nanopillars.** Theoretically calculated transmission spectra of the half-wave plate for linear co-polarized light (red) and cross-polarized light (blue). The axis of the wave-plate is at 45° to the axis of the input electric field. Solid lines: broadband metadevice with the dimensions a=285 nm, b=450 nm, c=860 nm and D=750 nm [see Fig. 2. in the main text for details]. Dashed lines: narrowband metadevice with the dimensions a=160 nm, b=470 nm, c=1080 nm and D=950 nm.

We now provide the details of calculations and numerical simulations that were used to produce the multipole analysis in the main text. To begin, we performed a fullwave numerical simulation of the silicon nanopillar array with *CST Mircowave Studio*, using experimental crystalline silicon permittivity data of Palik[9]. The internal electric field distribution $\mathbf{E}_{\text{CST}}$ of a single nanopillar was then exported from these simulations. Using the silicon permittivity, the exported internal field data could then be related to the corresponding internal polarization currents according to:

$$\mathbf{J}(\mathbf{r}) = -i\omega\left[\bar{\epsilon}(\mathbf{r}) - \epsilon_0\right]\mathbf{E}_{\text{CST}}(\mathbf{r}) \tag{24}$$

By using Green's functions, we were then able to calculate the scattered field produced by this internal polarization current distribution of the single silicon nanopillar, according to Equation (9). More specifically, we calculated the scattered electric field from the exported internal current distribution, when the surrounding environment is homogeneous free space, *i.e.* without either the substrate or other nanopillars in the array. We could subsequently project the scattered field on the surface of an arbitrary bounding sphere onto spherical multipoles, to thereby obtain a multipole decomposition as per Equation (2). Further details of the projection onto spherical multipoles be found in the work of Grahn *et al.*[2]. By following this procedure, we were able to obtain the spherical multipole decomposition of the scattered field from a single nanopillar in the array. However, as is discussed earlier in this section, the forward and backward scattering of single element of the array can be directly related to that of the array using Equation (21). Subsequently, the transmission and reflection associated with particular combinations of spherical multipoles could then be calculated from the backward scattered field of the array, as per Equations (22) and (23). Using this approach we were able to produce the plots used in Figure 2 of the main text.

C. EXAMPLES OF BOTH BROADBAND AND NARROWBAND METADEVICES

Here show that both broadband and narrowband operation can be achieved in Si nanopillars of similar sizes depending on their exact geometries. We design two half-waveplates with similar sizes both operating at around telecom wavelengths. And we calculate numerically the efficiency of 90° rotation of linear polarization. Figure S1 shows the corresponding spectra and lists the geometrical parameters.

If we exemplarily consider the bandwidth of the devices as the spectral region with contrast 100:1 between crossed- and co-polarized light, then we find that the broadband metasurface features 152 nm bandwidth, while the narrowband metasurface has 19 nm bandwidth, i.e. 8 times smaller.

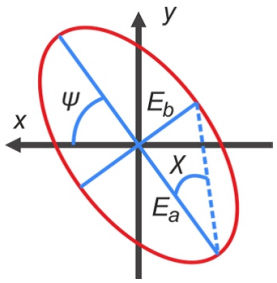


FIG. S2. Schematic of the polarization ellipse with the Stokes coefficients of ellipticity angle χ and polarization-inclination angle ψ. E_a and E_b are the main polarization axes (solid blue lines) of the polarization ellipse.

D. NANOFABRICATION PROCEDURE

For our sample fabrication, we employ a layer of poly-crystalline silicon (Poly-Si) created on a four-inch quartz wafer by using low-pressure chemical vapor deposition (LPCVD) in horizontal tube furnace . The thickness of the Si film is 850 nm, and the thickness of the substrate is 500 micro-meters. The pattern is written using a JEOL 9300FS 100kV electron beam lithography (EBL) tool. Polymethyl Methacrylate (950K PMMA A4) is used as the positive tone resist. A 10 nm thin layer of Cr is evaporated on top of the photoresist to prevent charging during the lithography. After the EBL process, wet chemistry removal of the Cr charge dissipation layer, and sequential development, one more 20nm thick Cr layer is e-beam evaporated and lifted off to create a hard mask on top of the poly-Si film. The resulting pattern is translated into the surface by means of anisotropic reactive ion etching (RIE) of the silicon layer not masked by chromium. Finally, the Cr mask is removed by wet etching.

E. EXPERIMENTAL SPECTRAL POLARIMETRY

To retrieve polarization states of light transmitted through the metadevices we employ Stokes vector polarimetry, and measuring 4 Stokes coefficients.

The Stokes coefficeints provide a complete description of the polarization state of light in terms of its total intensity I_{tot}, (fractional) degree of polarization ρ, polarization inclination angle ψ, and the ellipticity angle χ. The ellipticity $\tan(\chi)$ is defined as the ratio of the two axes of the polarization ellipse (see Fig. S2), and the polarization inclination is described by the angle between the main polarization axis and the x-axis of the laboratory coordinate system.

Experimentally, we find the Stokes coefficients by measuring the total intensity $I(\theta, \phi)$ where θ is the angle between the direction of the linear polarizers axis and the y-axis of our laboratory coordinate system. ϕ is the extra phase delay between the two orthogonal linear components of the electric field of the incident wave, that is introduced by the quarter-wave plate. The coefficients are then given by:

$$
\begin{aligned}
S_0 &= I(0^o, 0) + I(90^o, 0) = I_{tot} \\
S_1 &= I(0^o, 0) - I(90^o, 0) = \rho I_{tot} \cos 2\psi \cos 2\chi \\
S_2 &= I(45^o, 0) - I(135^o, 0) = \rho I_{tot} \sin 2\psi \cos 2\chi \\
S_3 &= I\left(45^o, \frac{\pi}{2}\right) - I(135^o, \frac{\pi}{2}) = \rho I_{tot} \sin 2\chi
\end{aligned}
$$

Figure S3 show exemplarily 6 transmission spectra $I(\theta, \phi)$ measured for the half-wave plate and 4 corresponding Stokes coefficients spectra retrieved from them.

In particular, spatially-varying polarization inclination angle ψ of the q-plate was measured with this method [See Fig. 5 in the main text].

Retardance (optical delay between two orthogonal polarization introduced by a wave-plate) was found as:

$$
\text{Retardance} = \frac{1}{4\pi}\left[\cos^{-1}\left(-\frac{S_2}{I \cdot \rho}\right) + \sin^{-1}\left(-\frac{S_3}{I \cdot \rho}\right)\right]
$$

We note that our retardance retrieval procedure is a simplified version of full Mueller's matrix polarimetry[10,11] which is applicable in our case since in the spectral region of interest total transmission intensity I and degree

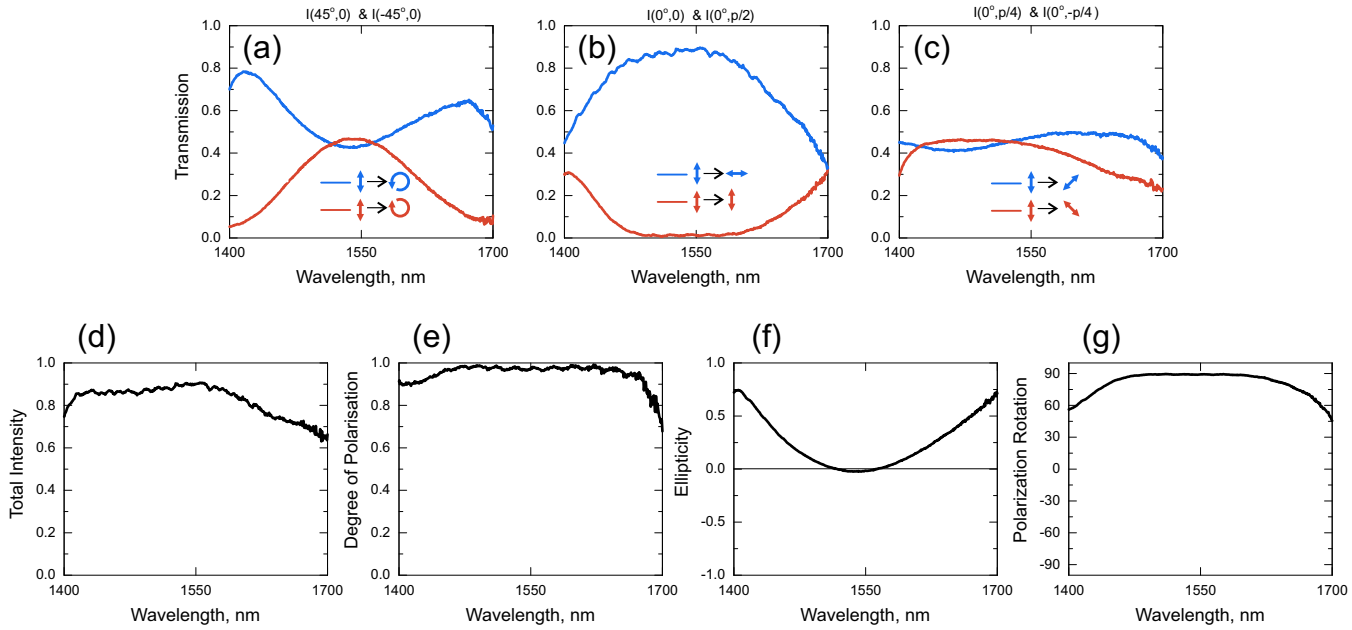


FIG. S3. Polarimetry of half-wave plate spectra. (a-c) 6 transmission spectra measured with 6 different analysers. (d) Total Intensity I, (e) Degree of Polarisation ρ, (f) ellipticity $\tan(\chi)$, (g) Polarization Inclination Angle ψ

of polarization ρ are not functions of incident polarization (e.g. our procedure works for transparent, non-scattering samples). This allows to retrieve the retardance from 6 experimenatal measurements versus 36 measurements required by the conventional method.

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